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Author(s): Elisabeth O. Selkirk

Source: *Linguistic Inquiry*, Summer, 1980, Vol. 11, No. 3 (Summer, 1980), pp. 563-605

Published by: The MIT Press

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Linguistic Inquiry Volume 11 Number 3 (Summer, 1980) 563-605.

Elisabeth O. Selkirk

The Role of Prosodic Categories in English Word Stress

1. Prosodic Categories and the Theory of Prosodic Structure

The principal aim of this article is to show that the proper representation of the patterns of stress in English words does not involve the phonological feature [±stress]. Such a feature forms part of almost all previous conceptions of the representation of English word stress in the framework of generative phonology,[1] in which rules are postulated for the assignment of the feature [±stress] to the vowels, or syllables, contained in words. It will be argued here that this feature and the rules for assigning it can, and should, be done away with in an analysis of English stress (thus paving the way for eliminating the feature entirely from the vocabulary of phonological theory).

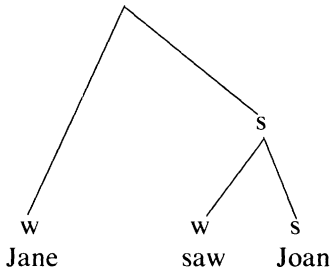
The theory of phonology in which the analysis to be proposed here is couched takes as basic the notion that a phonological representation has a hierarchically arranged suprasegmental organization, and it draws heavily on the fundamental insights into the workings of English stress made by Liberman (1975) and Liberman and Prince (1977) (hereafter LP). Central to the LP approach is the idea that the relative stress, or prominence, of elements of an utterance is to be explained as resulting from (i) the way in which the elements are grouped into constituents in a binary-branching "metrical" tree[2] and (ii) the relation strong/weak, or *s*/*w*, that is defined on the pairs of sister nodes comprising the tree. So instead of the standard theory numerical representation of stress as in $\overset{2}{Jane}\ \overset{3}{saw}\ \overset{1}{Joan}$, LP propose the representation (1).

[1] Cf. Chomsky and Halle (1968) (hereafter SPE), Halle and Keyser (1971), Halle (1973), and Liberman and Prince (1977), among others. It should also be noted that Schane's treatment of stress (1979), which employs the notation *s* and *w* to make stressed and stressless syllables, is but a notational variant of an analysis employing [+stress] and [−stress], for in the absence of a theory of hierarchical structure like Liberman and Prince's, the notations *s* and *w* are not designations of *relative* prominence, but are interpreted, like [±stress], as absolute features of the syllables themselves.

[2] The term *metrical* is put in quotations here, for it will not be adopted in the context of the theory being elaborated in this article. Indeed, the term *prosodic* will be employed in referring to the hierarchically arranged suprasegmental structure of the phonological representation. I have made this choice because the term *prosodic* is a traditional one for designating suprasegmental entities in linguistics, and because it seems more neutral than the term *metrical*, evoking more than the rhythmic arrangement of elements.

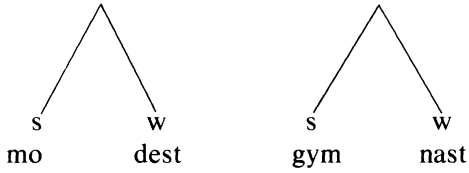
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(1)



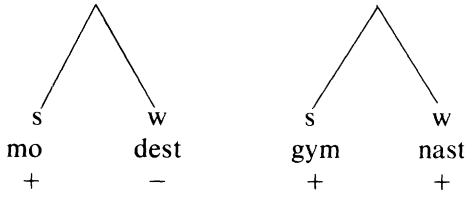
LP point out, however, that a representation consisting merely of a binary-branching tree with nodes labeled *s*/*w* is unable to capture all the necessary distinctions in prominence that one finds within the word in English. Consider the pair of words *módĕst* and *gýmnàst*. Both have primary stress on the first syllable. Given the LP approach, this means that the first syllable is *s*, and the following sister syllable is consequently *w*:

(2)



The suprasegmental structure assigned to both is identical. Yet the stress patterns of the two are not the same: the second syllable of *modest* is stressless, whereas that of *gymnast* has a secondary stress. In order to represent this distinction (as well as the comparable distinction shown by the pairs *tórmènt*/*dórmănt*, *máscòt*/*múskĕt*, *cónvìct*/*vérdĭct*), LP propose that the feature [±stress] be kept alive, and that it, in combination with the *s*/*w*-labeled binary-branching tree, be involved in representing the prominence patterns of words in English. Their revised representations for these words are as shown in (3):

(3)



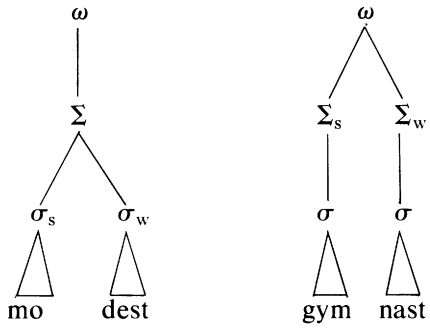
Thus, it is the syllables containing a vowel bearing the feature [+stress] which are interpreted phonetically as being stressed or prominent. The *s*/*w*-labeled tree then serves to define relations of relative prominence.

My approach to this matter is somewhat different. It involves an enrichment of the LP conception of suprasegmental phonological, or prosodic, structure, an enrichment which allows *all* aspects of the stress patterns of English words to be attributable to the

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characteristics of the higher order suprasegmental structure. This enrichment consists in positing the existence of a set of *prosodic categories*, which are isolable subunits of the prosodic structure, in associating "labels" corresponding to these categories with the appropriate nodes in the tree, and in allowing for the possibility that nodes with different prosodic category labels may be interpreted differently by processes of the phonology. In the case of English word stress, I propose that the prosodic categories *syllable* (σ), *stress foot* (Σ),[3] and *prosodic word* (ω) have a fundamental role to play. In particular, the presence of prosodic category labels in phonological representation permits a straightforward solution to the problem of *modest* vs. *gymnast*, which have representations (roughly) as shown in (4) in this theory:

(4)



The solution relies on the idea that a syllable which *is a* stress foot will never be interpreted as a weak (i.e. stressless nonprominent) syllable, even when the Σ which (exhaustively) dominates it is weak. That is, being a Σ always implies some degree of prominence. Thus, the *-nast* of *gymnast* has a "secondary stress". The syllable *-dest* of *modest*, on the other hand, is not a stress foot itself. It is merely a weak (*w*-labeled) syllable, here contained within a Σ, and as such it is interpreted as stressless, or nonprominent, and is subject to reduction. According to this theory, then, it is the organization of syllables into stress feet which provides the explanation for the *modest-gymnast* difference, and, more specifically, the presence of the category label Σ in the representation which allows for the difference in phonetic interpretation.

The category stress foot does more than permit a solution to the *modest-gymnast* problem, however. It has a broader function than that of setting up a label in phonological representation which may be identified and assigned a distinctive interpretation. As we will see in the following section, it is essentially in terms of this category of prosodic structure that all the generalizations concerning the patterns of *stress placement* in English words are to be captured.

In the LP system, the placement or location of stress is determined through the

[3] The term *stress foot* has been chosen rather than *foot*, in order that the prosodic unit at issue be clearly distinguished from the span of the utterance referred to as the foot by Abercrombie (1964) and others working in that vein.

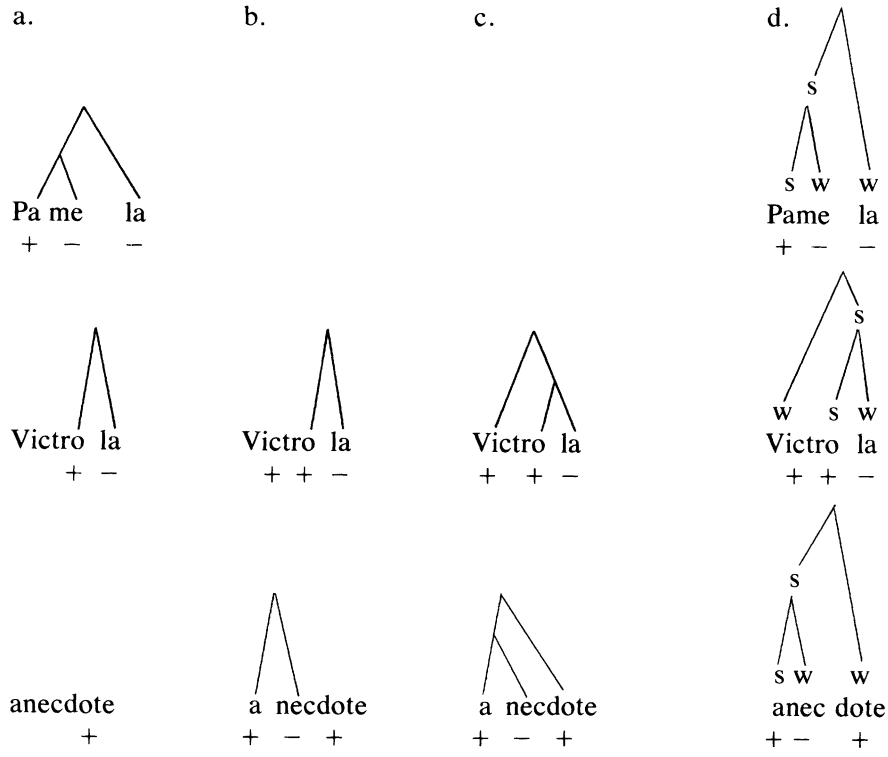
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application of a standard type of rule which assigns the feature [±stress] to the vowels of a word. (5) is a simplified version of the stress rule LP adopt.

(5) *English Stress Rule* (a standard version)
[+syll] → [+stress] / [. . . ___ C_0(V(C)) (V C_0)]

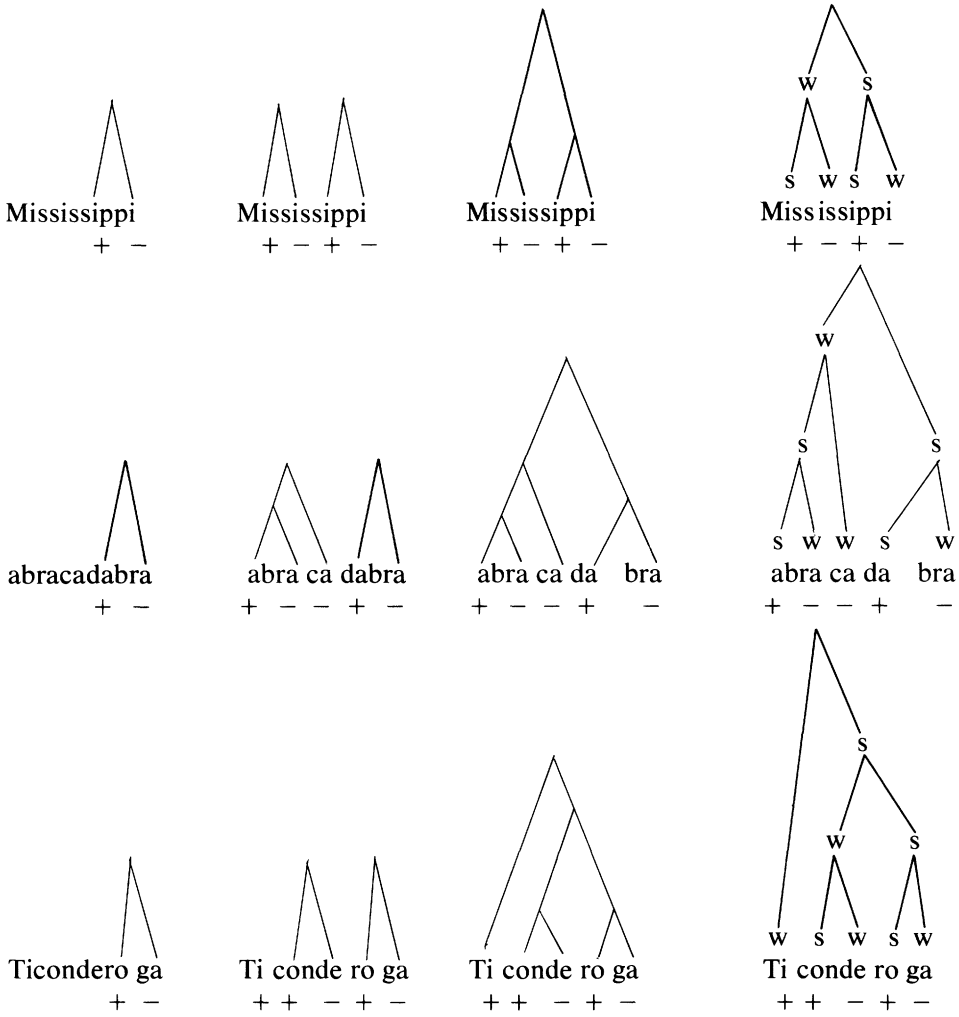
(The syllable is given no role in this version of the rule, though LP (p. 272) acknowledge that syllabic structure must be taken into account in a definitive treatment of stress placement in English words.[4]) Concomitant with the (iterative) assignment of the stress feature to vowels is the building of a "metrical" structure. With each iteration, from right to left, of (5), a left-branching tree is built joining the syllable with the [+stress] vowel to the following syllable(s) with a [−stress] vowel. In column (a) of (6) we see the results of a single application of the rule.

(6)



[4] The importance of the division of a word into syllables in an insightful characterization of English stress patterns has been underlined by Hoard (1971), Anderson and Jones (1974), and Kahn (1976), among others.

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In column (b) we see the results of a second and, in the case of *Ticonderoga*, third application of the rule. LP refer to the branching structure which immediately dominates the sequence of [+stress] syllables followed by [−stress] syllables as a *foot*. (Foot structure is built only when the newly assigned "+" is followed by a "−".) Also attendant upon this leftward iteration of the stress rule is the construction of a right-branching higher order tree which links together the feet and/or lone [+stress] syllables. In column (c) we see the examples with their trees thus completed. The highest node of the tree is referred to as the *mot*. Finally, the determination of the strong/weak relations in the tree is made by the LP Lexical Category Prominence Rule (LCPR), (7). (Cf. column (d) of (6).)

(7) *LCPR*

Given a pair of nodes $[N_1\ N_2]$, N_2 is *s* iff it branches.

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This, in a nutshell, is the (much simplified) LP analysis of English word stress: stress placement and tree-building are accomplished by the stress rule (5), the *s/w* labels are given by (7).

The LP *foot* has no special role to play as a structural unit in this analysis. It is merely a term informally used to designate the subtrees created as the stress rule applies; reference to the foot is not made in any other part of the LP analysis. The "foot" is not a primitive notion of the LP theory. Nor is the syllable. Indeed, prosodic categories have no role in the LP analysis of English word stress, and it is in this way that the approach being advocated here differs radically from theirs. Here, the syllable, the stress foot, and the prosodic word do not have a derivative status, but are instead the basic constitutive units of prosodic structure in terms of which the stress generalizations are expressed.

(It is the case that in their treatment of compound stress and the *mot* (which is the same as the *prosodic word*), LP foreshadow in a certain way the theory of prosodic categories and category-labeled prosodic structure being developed here. The reader will recall LP's argument (pp. 269–270) to the effect that, in applying to compounds, the LCPR must "know" whether or not some substructure of the tree is a *mot* or not, for the branching within a *mot* does not "count" in determining the *s/w* patterns in compounds. One could interpret LP's statement as making the claim that the appropriate nodes of the tree must therefore be labeled *mot*, and that the *mot* is thus a prosodic category having a direct role in the analysis of supraword stress in English.)

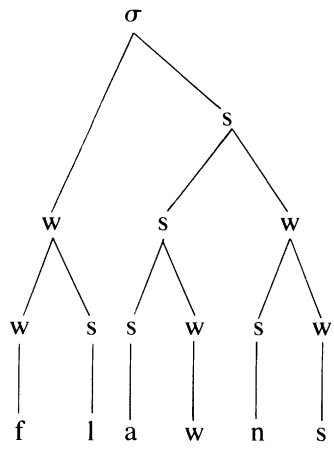
Given the theory of the representation of word stress I am proposing, the analysis of the stress system of any language, and of English in particular, consists of two things. First, in defining for that language a set of *prosodic well-formedness conditions*, which specify how well-formed syllables, stress feet, and prosodic words are constituted. Second, in specifying the *syntactic domain* within which these conditions obtain, that is, within which the segments of a phonological representation are grouped into well-formed syllables, the syllables into well-formed stress feet, and the stress feet into well-formed prosodic words. In general, it will be said that for any prosodic category K_i, be it above or below the level of prosodic word, a triplet $K_i = (C_{K_i}, P_{K_i}, D_{K_i})$ is to be defined, where C_{K_i} are the *principles of construction* specifying the nature of structure internal to K_i (constituency in terms of other categories, direction of branching, etc.), P_{K_i} are the *principles of prominence* specifying the *s/w* relations of the subtrees constituting K_i, and D_{K_i} is the *syntactic domain* within which these well-formedness conditions are met. In what follows, the hypothesis will be entertained that these well-formedness conditions are satisfied at underlying phonological representation, which is to say that prosodic structure is "established" or defined from the start in a phonological derivation (which, given our approach, is tantamount to saying that "stress" is the "first rule" to "apply" in the phonology).

It will not be possible within the context of this article to discuss the English syllable in any detail—these details are not crucial to the analysis of English stress to be offered. But to give an idea of the analysis of the English syllable that will be assumed

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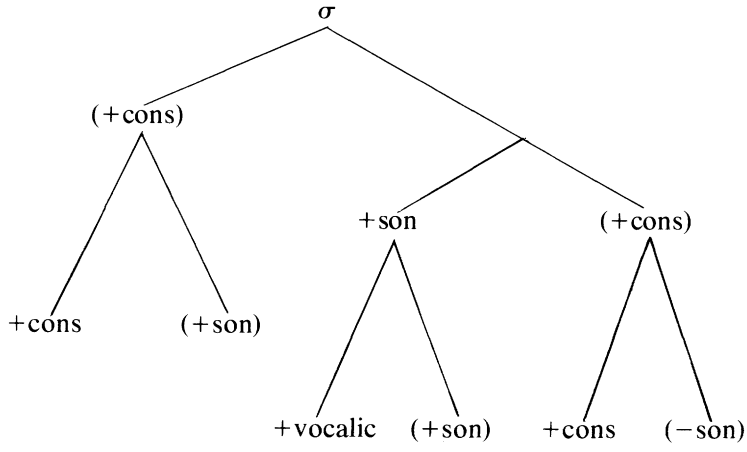
(one which is developed at length in Selkirk (forthcoming)), a few remarks would be useful. It can be argued that the syllable in English has two major constituents—the onset (the initial consonant cluster) and the rhyme—and that the rhyme itself has a bipartite structure, dividing into the nucleus and the coda.[5] The *s/w* relations between the constituents would seem to correlate with the degree of relative sonority. Accordingly, a complex monosyllable like *flounce* [flawns] has the structure given in (8).

(8)



In Selkirk (forthcoming) I suggest that the principles of construction of the syllable, C_{σ}, be expressed in part by a template, roughly of the form (9), which characterizes the gross features of syllable structure in the language. (The template could also include specification of the prominence relations.)

(9) *English Syllable Template*[6]



[5] I adopt here what is essentially the analysis of Fudge (1969), following in its general lines what is advocated in Pike and Pike (1947).

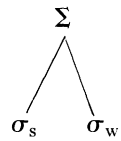
[6] This version of the syllable template for English is different in the details of onset and coda composition from the one proposed by Halle and Vergnaud (1979). Unfortunately, I cannot now evaluate those differences, irrelevant to the basic issues at hand.

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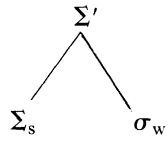
The important thing to bear in mind is that the syllable has an internal structure, hierarchically arranged.

It is the prosodic category stress foot, mediating between the syllable and the prosodic word, which is central in the analysis of English word stress. Stress feet are of two basic types—monosyllabic and bisyllabic:

(10) a. 

b. 

The two syllables of the bipartite foot are always in a *s/w* pattern. An additional type, the stress superfoot, may consist of a foot (*s*) followed by another syllable (*w*):

(10) c. 

The weak syllables of Σ and Σ′ are the ones interpreted as "stressless". "Stressed" syllables are either the strong of a bisyllabic Σ, or the sole syllable of a monosyllabic Σ. It should be clear, then, that "stress placement"—the patterns of stressed vs. stressless syllables—follows immediately from the way a word is constituted in terms of stress feet (and superfeet).

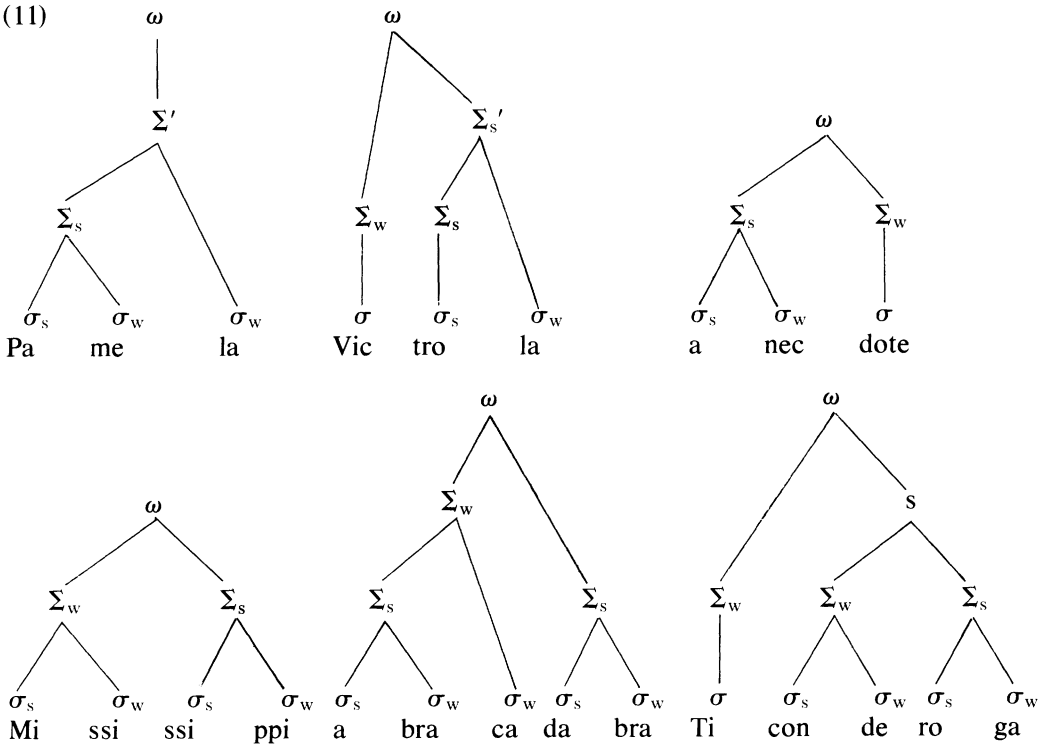
The category prosodic word is defined as being constituted of a sequence of one or more Σ or Σ′ joined in a right-branching structure (= the principles of construction, C_ω). The Lexical Category Prominence Rule (7) specifies the *s/w* relations in the tree above the level of Σ or Σ′. That is, the LCPR represents the principles of prominence, P_ω, that are proper to the prosodic word. The stress foot, and superfoot, have their own prominence principles.

The syllable, the stress foot, and the prosodic word all have the same syntactic domain in English: the simple (nonbranching) *stem*. The stem is a level of morphological structure in English smaller than the syntactic word. (It is argued for in Selkirk (forthcoming).) Suffice it to say here that the simple stem does not include the so-called neutral affixes. This means that neutral affixes are "outside" the domains of syllabification, stress foot formation, and prosodic word formation; that is, they are neutral with respect to stress. More will not be said on this question, not a central one, at present.

To come to a first understanding of this new approach to stress, which is to be elaborated in section 2, let us examine the examples of well-formed prosodic words used earlier on in (6):

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(11)



(I have taken the convenience of not representing syllable-internal branching in these representations.) *Pamela* is a word consisting of a single superfoot, the initial Σ of which is bisyllabic. Within this prosodic word, the *s/w* relations of the syllables are determined by the principles, P_Σ, proper to the stress foot (and superfoot), not by the LCPR. *Victrola* is parsed as a sequence of a monosyllabic Σ followed by a Σ′ whose inner Σ is monosyllabic. The branching of the Σ′ merits it an *s*, by the LCPR. *Anecdote* is analyzed as consisting of two Σ, the first bisyllabic, the second monosyllabic. Because the second Σ does not branch, the LCPR assigns *s* to the first. *Mississippi* and *abracadabra* are also prosodic words consisting of a sequence of just two Σ (or Σ′), of various sorts. Since in both instances the righthand Σ is bisyllabic and thus branching, the LCPR assigns an *s* to it. *Ticonderoga* is the sole example seen so far of a ω with more than two Σ, thus requiring some more complicated organization of the stress feet with respect to each other: the last two are joined together, and this subtree is joined with the first. The LCPR consistently assigns the *s* prominence to the right since all right nodes are branching here.

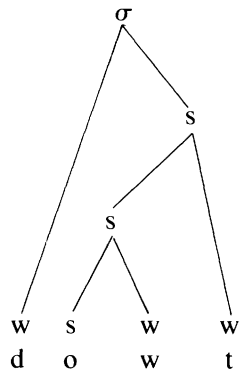
The stress foot structures of (10) provide only the gross outlines of the shape of possible stress feet in English. As will become clear in section 2, where the principles of construction and prominence for the stress foot are fully spelled out (in the form of

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templates), there are severe restrictions on what sorts of syllables may occupy the various positions within the foot. It is generally the case, for example, that an open syllable containing a lax vowel may not be a stress foot on its own, whereas a syllable containing a tense vowel must always be either a stress foot or the strong of a Σ (cf. the *Ti* and the *ro* of *Ticonderoga*). A lax closed syllable has a much freer distribution. It may be a foot on its own, as with the first syllable of *Victrola* or the second syllable of *gymnast* (as well as the first) (cf. (4)), or it may be part of a bipartite foot, as with the second syllable of *modest* or of the Σ *anec-* of *anecdote*. The constitution of a prosodic word in terms of syllable types will determine to some extent how that word is to be parsed into stress feet, though, as we have seen, certain options are available. In order for a prosodic word to be ruled well-formed, it must only be that there exists for it some one parsing into possible feet.

A couple of general remarks on the theory of prosodic categories are occasioned by the examples of (11). First, it is important to note that the branching inside a syllable, and, in particular, the branching of a rhyme, does not "count" from the point of view of the LCPR,[7] though branching inside Σ or Σ′ does. Were it to be relevant, the heavy syllable *-dote* [dowt] of *anecdote*, whose structure is as shown in (12) according to template (9), would have drawn the *s* assigned by the LCPR.

(12)



Thus, in the formulation of the LCPR it must be specified that the node variables N_1 and N_2 not stand for "σ", i.e. that "$N_1, N_2 \neq \sigma$". Which means that the LCPR has to "know" which part of the prosodic structure tree is a syllable and which is not. Which in turn means that the prosodic category label "σ" needs to be present in the representation. This is of course what I am arguing for. (Note that a consistent version of the LP analysis of stress would also require that "σ" label the tree.)

A second point must be made concerning the labeling of a prosodic structure tree. As is evident from the representation of *Ticonderoga*, not all nodes of prosodic structure

[7] It should be noted that the branching of a syllable into onset and rhyme never seems prosodically relevant. That is, a syllable composed only of V does not seem to differ from one constituted of CV (or CCV, etc.). It is only the rhyme's branching which appears to make a difference.

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will have prosodic category labels. (The same point could have been made on the basis of the representation of the internal structure of syllables, where there are no category labels.) This is simply because not all nodes correspond to some prosodic category, i.e. to some distinctive unit of prosodic structure having its own principles of construction and prominence. The un-category-labeled node dominating the last two feet of *Ticonderoga* has no independent status: it exists only in virtue of the prosodic word, whose feet, when greater in number than two, are organized in right-branching fashion. As for the subconstituents of the syllable, they, too, have no independent status. The coda constituent [+cons (−son)], for example, has no independent privileges of distribution. It is to be found only in second position within the rhyme of the syllable. This situation is to be contrasted with that of the stress foot or the syllable. The prosodic word is composed of any number of stress feet, of any of the various sorts, freely combined. The stress foot is thus a constitutive unit requiring an independent definition; that is, it is a category of prosodic structure. As for the prosodic category syllable, it too is a building block, one in terms of which the stress feet of various sorts are defined. As we will see, the distribution of the different syllabic types within stress feet is not so free as that of the feet within the prosodic word, but these restrictions in no way vitiate the status of the syllable as an independently defined unit of prosodic structure. To sum up, this theory of phonological representation allows for the existence of nodes without category labels. The analogy that can, and will, be drawn between syntactic representation and the conception of phonological representation being developed here is thus not complete.

Let us now review the structure of the argument being made against the LP approach to English word stress. In what has preceded I have claimed that all the generalizations about English word stress can be neatly captured by an analysis couched in the theoretical framework outlined here, which, among other things, gives no place to [±stress]. (This claim will be substantiated in section 2, where it will also be argued that my approach makes a real improvement in capturing those generalizations.) The further claim can be supported (and will be, directly below) that the (innovative) features of the theory being proposed, upon which the analysis of English stress crucially relies, have independent motivation. I am speaking of the definition of well-formedness conditions for prosodic categories, the association of labels for those categories with nodes of the prosodic structure tree, and the availability of those labels to interpretation. What this means is that any descriptively adequate theory of prosodic structure must include those features I propose. This being the case, there is no reason for the feature [±stress] to be included as well in the vocabulary of the theory of phonology. [±stress] and rules for assigning it are superfluous, and the theory which does not make use of them must be adopted, for it is simpler and streamlined, capturing elegantly, with the minimum of devices, the essential properties of prosodic structure and its relation to rules of the phonology.

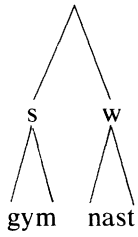
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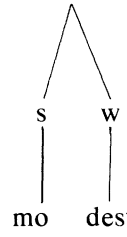
scribed, I would like to briefly compare this approach, in its general features, to another different approach to prosodic structure and the analysis of stress systems, one which is also a development of the LP framework that manages to do away with the feature [±stress]. The approach is that of Halle and Vergnaud (1979), adopted also by McCarthy (1979), and I would like to argue for the superiority of the theory I am elaborating over theirs. It is far beyond the scope of this article to deal with the wealth of material treated by Halle and Vergnaud (hereafter HV), so the argument will not be based on providing the appropriate superior analyses within the framework developed here. Rather, it will treat more general problems.

The Halle–Vergnaud approach is similar to mine in many respects. One of their projects is to do away with the segmental feature [±stress] in phonological theory, and to do so by putting the focus on the properties of prosodic structure and in particular on the prosodic categories syllable and foot as primitives of accentual systems. Their analyses of the stress patterns of a variety of languages rely, in part, on defining what constitutes a possible syllable and a possible foot in those languages. Yet the conception of phonological representation they adopt makes no place for prosodic category labels like Σ and σ; it is merely an LP type of representation, consisting of a binary-branching *s/w*-labeled tree. How is it, then, that HV could characterize the difference between *gymnast* and *modest* in English? HV do not in fact undertake an analysis of English stress, but, extrapolating from their approach to not dissimilar problems in other languages, one can construct an analysis such as they might provide. It would amount to saying that, in general, a syllable with a branching rhyme (that is, one with either a tense (long) vowel, or one that is closed with a consonant) is interpreted as being stressed. *Gymnast* would be viewed as having the structure in (13a).[8]

(13) a.



b.



The second syllable of *modest* could not be treated in this way, however. What HV could say, and they have this possibility within their theory, is that the branching of the rhyme in *-dest* is not "projected". That is to say that the branching (which is indeed there, *-dest* being a heavy syllable) is idiosyncratically not available to phonetic interpretation, and hence that *-dest* is treated on a par with light, i.e. open, lax, syllables. *Modest* would have the representation in (13b); since the "nonbranching" *-dest* is a *w* with respect to its sister *s*, it is interpreted as stressless. With these examples the most

[8] In the HV system, the branching into onset and rhyme of a CV syllable is never "projected", i.e. made relevant. Cf. footnote 7.

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fundamental difference between the HV approach and mine emerges: according to HV, the interpretation of a syllable as stressed or stressless relies only on the abstract branching properties and *s/w* relations of the representation (some of which may or may not be "projected"); according to my approach, the interpretation depends as well on the labeling of the nodes, on whether or not some subtree is an instance of a particular prosodic category.

The argument for my treatment of the stress phenomena in English and against an HV type of treatment is based first of all on the claim that the conception that it presupposes of the nature of prosodic structure as being category-labeled and of the manner of its interpretation is independently necessary, being motivated by a wide range of evidence from phonology. (Some of this evidence will be reviewed directly below.) The labels corresponding to the prosodic categories syllable, stress foot, and prosodic word must form part of the phonological representation, and they are available to interpretation by processes of the phonology. The second claim is that my treatment is the best one, given the devices this prosodic structure theory makes available. It is equally true that the branching internal to syllables and their rhymes is independently motivated. Within prosodic structure, this branching must therefore be represented. So one could argue, with HV, that rhyme-branching is indeed what is at issue in the interpretation of English heavy syllables, and that the Σ label, though present in representations, is irrelevant. This argument is extremely weak in the case of English, however, given that not all syllables which do have branching rhymes (as with the *-dest* of *modest*) could be treated as such. Which is to say that branching becomes an ad hoc "diacritic" device. No such inconsistency arises with the analysis which depends on the stress foot label: all Σ are interpreted in the same way. A spade is a spade. The Σ has a reliable relation to prominence, and so clearly the analysis based on it is the more desirable one.

It is a fairly natural move to consider that prosodic category labels form part of a representation, once prosodic categories have been given the role of significant units of linguistic description. An analogy to syntax should make this clear. In the syntactic description of most, possibly all, languages, a constituent unit *noun phrase* must be isolated, in order to account for the recurrent patterns of distribution in the sentence of words belonging to such classes as noun, adjective, determiner, quantifier, etc. In other words, the unit *noun phrase* is a necessary element of a description of the *phrase structure* of sentences in a language. And, given the theory of syntactic representation, the label NP corresponding to that unit is associated with the node dominating that subtree which *is a* noun phrase. In the case of NP, various sorts of evidence do indeed show that the label *must* form part of the representation. Transformations apply selectively to subtrees of a syntactic representation depending on whether they are instances of the category NP rather than VP, AP, etc.; rules assigning a semantic interpretation to the representation may be specific to categories such as NP; and rules of the phonology may pick out NP to the exclusion of other categories. (See Clark (1978) on

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the tonal properties of the noun phrase in Igbo, for example.) Thus, processes which "look at" a syntactic representation show that the label corresponding to the phrase structure unit noun phrase must be there. What we see here, and in general, is a systematic convergence: those units of syntactic description motivated on distributional (phrase structure) grounds are just those which may be picked out by grammatical processes which apply to a syntactic representation. The labeling of the syntactic representation is the theoretical device which makes this possible, which predicts this convergence. In similar fashion, in phonology a case can be made for the association of prosodic category labels with nodes of the hierarchically arranged phonological representation. Those units of phonological description which are motivated on distributional, i.e. phonotactic, grounds (that is, those which have their own principles of construction and prominence) are just those which play a role in processes, accentual or nonaccentual in character, which look at or apply in terms of the phonological representation. This convergence can be demonstrated for all levels of prosodic categories: the syllable (σ), the stress foot (Σ), the prosodic word (ω), the phonological phrase (φ), the intonational phrase (I), and the utterance (U) (cf. Selkirk (1978a; 1978b to appear) and especially Selkirk (forthcoming)).

Rules which "look at" the phonological representation, and, in particular, at prosodic structure, fall into a variety of classes. These include (i) certain *principles of prominence* (perhaps not all), which, like the Lexical Category Prominence Rule, assign strong/weak relations in function of the configurations of prosodic structure, (ii) *prosodic structure transformations* such as Resyllabification or Defooting (see below), whose explicit function is to modify prosodic structure relations (but not the segmental composition) of the representation, (iii) *rules of segmental phonology*, whose characteristic domains of application are defined in terms of the constituents of prosodic structure, and (iv) *rules of phonetic interpretation*, such as those determining the relative duration of units of the representation, whose effects, though not "phonemic", or even potentially distinctive, may be specific to some particular language. Rules of all four types bear witness to the presence of prosodic categories in phonological representation. In the brief review that follows, the wealth of evidence provided by such processes will only be hinted at. (For a fuller investigation of the dependence of phonological processes on prosodic structure, the reader is referred to Selkirk (forthcoming) and the references cited therein.)

The necessity of prosodic category labels in the representation is particularly obvious in the case of the syllable, perhaps the best understood of prosodic categories. It was already mentioned above that the English LCPR must "know" which subtrees of prosodic structure correspond to syllables, in order that the branching internal to syllables (and their rhymes) be removed from consideration in the assignment of *s/w* prominence relations. As for rules of the segmental phonology that are sensitive to syllable structure, there are many which are by now well known. A classic case is the rule aspirating voiceless stops in English. As numerous scholars (Jones (1956), Anderson

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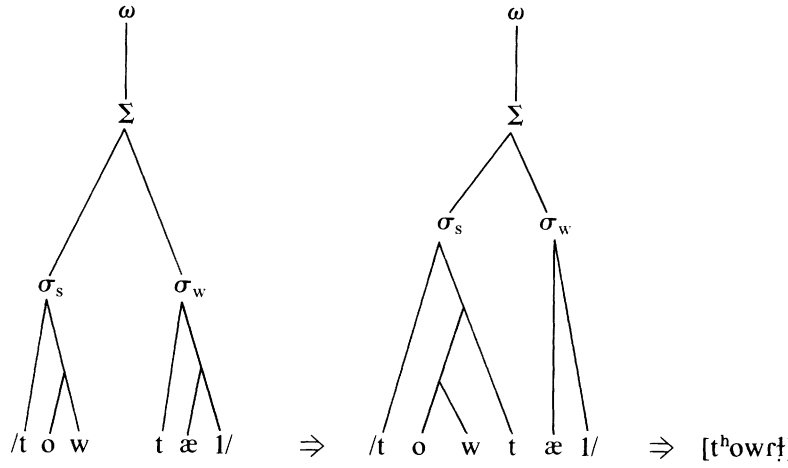
and Jones (1974), Hoard (1971), and Kahn (1976; to appear) among them) have taken pains to demonstrate, the generalization is that syllable-initial voiceless stops, and no others, are subject to aspiration. (The contrasts *pat* [p^{h}æt] vs. *spat* [spæt] vs. *sap* [sæp] help illustrate this well-known point.) The English aspiration rule must thus be formulated as shown in (14):

$$(14)\quad \begin{bmatrix} -\text{cont} \\ -\text{son} \\ -\text{voice} \end{bmatrix} \rightarrow [+\text{aspirated}] \,/\, {}_{\sigma}(\,\underline{\quad}\,\ldots)_{\sigma}$$

Additional well-known examples of syllable-sensitive rules are provided by the rule of syllable-final obstruent devoicing in German (cf. Vennemann (1972)) or the rule of French converting a tautosyllabic vowel plus nasal combination into a nasal vowel (cf. Basbøll (1978)). Perhaps less well known, but equally telling, is the example of the process found in Colloquial Egyptian Arabic which "spreads" the feature of pharyngealization from a pharyngealized phoneme to every other segment—vowel or consonant—that is *contained within the same syllable* (cf. Lehn (1963), Broselow (1976)). I leave it as an exercise for the reader to verify for herself that labeling the syllable subtrees of the representation with "σ", and thereby allowing for the formulation of rules in terms of that "labeled bracketing", as in (14), represents the only simple and straightforward means of capturing these generalizations. Finally, research of an experimental phonetic character has demonstrated, for example, that the syllable in English is a unit of timing, with an "ideal" overall duration (the "ideal" may vary depending on the broader context), such that the derivations of the segments making up the syllable are negatively correlated (cf. Lehiste (1971)).

As for the stress foot "Σ", it can be argued, for example, that it is the domain of a prosodic structure transformation of Resyllabification which "moves" the onset consonant of a weak syllable in the foot to the coda position of the preceding syllable, as in the derivation of *total*, cf. (15).

(15)



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This transformation forms part of a reanalysis of "interludes", the so-called ambisyllabic consonants of English, which is elaborated in Selkirk (forthcoming): the lack of aspiration of voiceless stops in this type of medial position, and the flapping of *t* and *d*, is attributed (in part) to their *syllable-final* position in derived structure. Kiparsky (1979) has a different solution to the problem posed by the phonological behavior of these consonants, but one which nevertheless makes crucial use of the stress foot as the domain of the rules involved. This is not the place to argue the relative merits of the two solutions; I will simply point out that in either case the Σ must form part of the phonological representation. The Σ can be shown to be the characteristic domain of rules in other languages as well. In Selkirk (1978a) (as well as in Selkirk (forthcoming)) I argue that in French a stress foot is constituted either of a single syllable or of two, where the second, weak, syllable contains a schwa. The various processes of schwa deletion in the language (see Dell (1973) for the authoritative description of these) are then all argued to be the same: a schwa deletes only when the syllable containing it is the second member of a foot. In that same article a reanalysis of the French rule of "closed syllable adjustment" is offered which solves certain classic puzzles dealt with by Selkirk (1972), Dell (1973), and Basbøll (forthcoming). It is argued that the proper domain of the rule at issue, which converts /e/ to [ɛ], cf. *célébrer* [selebre] ~ *célèbre* [selɛbr(ə)], and /ə/ to [ɛ], cf. *sevrer* [səvre] ~ *sèvre* [sɛvr(ə)], is the stress foot:

$$e, ə \rightarrow ɛ \;/\; {}_{\Sigma}(X ____ Y)_{\Sigma}$$
$$\text{where } Y \neq \phi$$

Turning next to the question of the phonetic interpretation of Σ, recall that it was suggested earlier that some prominence is always associated with it, or, in other words, that every Σ is "stressed". In the case of English, this means that the lone syllable of a monosyllabic stress foot is "stressed", and not subject to rules of vowel reduction and so on, and that the strong syllable within a polysyllabic stress foot or superfoot is "stressed". This notion that a Σ is always in some way prominent may provide some insight into the characterization of "stress-timed" vs. "syllable-timed" languages. I would suggest that syllable-timed languages, such as French, are those in which the Σ is essentially monosyllabic, which means that each syllable (= Σ) emerges as somewhat prominent. (The (relatively rare) instances of bisyllabic Σ in French have as their second member a schwa syllable, which is notoriously subject to reduction (by V deletion), so that on the surface the realization of these feet is (frequently) monosyllabic.) Stress-timed languages, on the other hand, are those in which the predominance of polysyllabic feet makes for a systematic difference in prominence among syllables in sequence, the weak syllable(s) in the foot taking a back seat to the strong. Now, if we think of the timing involved as being the (idealized) isochrony of Σ in sequence, then the rhythm of French is an almost invariable staccato succession of syllables, while the rhythm of English is nuanced by the variable composition of Σ, and the difference in prominence of the syllables within them.

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There is ample evidence that the prosodic word "ω" as well must be identifiable by processes that apply to the prosodic structure representation. Consider again the functioning of the LCPR in English. As was said above, LP themselves make the argument that this principle of prominence must "know" the breakdown of the utterance into units (which they call *mots*) corresponding to the prosodic word. And examples aplenty are available among rules of segmental phonology, many of which have ω as their domain. In English, the rule tensing nonlow vowels (cf. SPE), as in *potato* or *spaghetti*, applies at the end of the ω domain.

(16) $\begin{bmatrix} +\text{vocalic} \\ -\text{low} \end{bmatrix} \rightarrow [+\text{tense}] \,/\, {}_{\omega}(\ldots \underline{\quad})_{\omega}$

In French, ω-final position is the context for the rule of consonant truncation (cf. Selkirk (forthcoming)), just as in Sanskrit it is the context for cluster reduction and final devoicing and deaspiration (cf. Selkirk (to appear)). The ω also defines a domain *within or across which* (but not necessarily at the limits of which) a segmental rule may apply, as in the case of *ruki* and *nati* rules of classical Sanskrit:

(17) *Ruki*

$$s \rightarrow \text{ṣ} \,/\, {}_{\omega}(\ldots \begin{Bmatrix} r \\ u \\ k \\ i \end{Bmatrix} \underline{\quad} [+\text{seg}]_{a} \ldots)_{\omega}$$

(where $a \neq r$ or ṛ)

Nati

$$n \rightarrow \text{ṇ} \,/\, {}_{\omega}(\ldots \begin{Bmatrix} s \\ r \\ \text{ṛ} \end{Bmatrix} Q \underline{\quad} [+\text{son}] \ldots)_{\omega}$$

(where $Q \neq \ldots [+\text{coronal}] \ldots$)

This analysis of the domain of these well-known rules is discussed in Selkirk (to appear). Other segmental rules which have ω as the domain across which they apply are voicing assimilation of obstruents in Czech (Jakobson (1949)), vowel harmony in French (cf. Selkirk (1972), Dell (1973)), obligatory nasal assimilation in Catalan (cf. Moscaró (1976)), etc. Finally, experimental analysis of speech production in English has shown that the ω is assigned a distinctive phonetic interpretation (in terms of duration), one which allows the monosyllabic ω to be distinguished from a monosyllabic Σ and from the (strong) syllable that is first in a bisyllabic foot (cf. Nakatani and Schaffer (1978)).

What this brief review has shown is that the subtrees corresponding to the prosodic categories syllable, stress foot, and prosodic word must stand out and be identifiable as such in the representation, for a host of phonological processes appeal to information about the make-up of the representation in terms of these categories. The most natural

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device permitting this identification is the labeling of nodes, and this device is thus incorporated as part of the theory of prosodic structure. (Note that it is a consequence of this theory that *un*labeled nodes, those not corresponding to some prosodic category, may not serve as the characteristic domain of some phonological rule. So far as I know, this prediction is true.)

It should also be evident from the foregoing that a new conception of the relation between segmental and suprasegmental phonology is now available. These two are tightly linked, given this theory, for it is claimed (with supporting evidence) that the proper characterization of the domains for the application of segmental rules is to be made in terms of the suprasegmental prosodic structure. I would like to make the further claim that the domains of rules of segmental phonology are never to be characterized in terms of the boundary elements #, ##, etc., of the standard theory. These are supplanted entirely by the notion of prosodic structure domains, and thus should be eliminated as a superfluous device of the theory. The information about the surface structure of a sentence which was encoded in boundaries (cf. SPE, Selkirk (1972; 1974)) is, in the present framework, encoded in the prosodic structure itself, each category K_i of that structure being defined as well-formed relative only to some specific stretch of the syntactic surface structure (the syntactic domain D_{K_i}). There is not room to argue for the further claim here (this is done in Selkirk (1978b) and, at length, in Selkirk (forthcoming)). I mention it here in order to indicate yet another of the beneficial results of the theory of prosodic structure being laid out in this article: the conception of the mapping between syntax and phonology is an eminently simple one, consisting merely in characterizing the relation $M_{s/p}$ between the hierarchical representation of syntactic structure and the hierarchical representation of phonological (prosodic) structure.[9] (That relation is embodied in the set of triplets which define each prosodic category: $M_{s/p} = \{K_i/K_i = (C_{K_i}, P_{K_i}, D_{K_i})\}$.) Once established, it is the hierarchical prosodic structure, along with the string of sound segments, which governs the application of the phonological rules. Neither boundaries, nor the labeled bracketing of the syntax, are given any role to play in the formulation of rules in phonology.

2. A Prosodic Structure Analysis of English Word Stress

The analysis of English word stress that will be presented here incorporates many features of the LP analysis: the notion that prosodic structure consists of a binary-branching tree whose nodes are in the strong-weak relation; the notion that there exists an entity such as the prosodic word (their *mot*); the notion that within the prosodic word there exist subtrees corresponding to the placement of stress (these we have elevated

[9] This notion that the mapping is to be defined between the "grammatical hierarchy" and the "phonological hierarchy" is one entertained, but not extensively developed, by Pike (cf. Pike (1967)).

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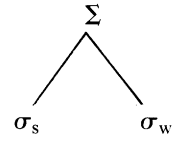
to the status of *stress foot*); the notion that these subtrees are themselves grouped into a right-branching structure upon which the *s/w* prominence relations are defined which determine, ultimately, the location of primary stress within the prosodic word. The spirit of the LP approach is that stress patterns are a matter of the prosodic hierarchy. My approach, which is to develop the conception of prosodic structure in postulating additional prosodic categories and thus eliminate the feature [±stress], permits an analysis of English where stress is entirely a matter of the prosodic hierarchy. The spirit of LP, I dare say, is thus carried to its logical conclusion.

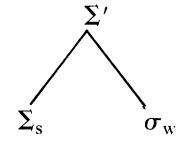
In what follows, I will first lay out the new core of the analysis, which consists essentially in defining the well-formedness conditions for the *stress foot* in English. These will take the form of a set of templates, as sketched in the preceding section. The empirical predictions made by these templates, somewhat different from those made by the standard stress-rule type of analysis, which LP adopt, will be investigated.

Because certain options in the parsing of a word (on the stem domain) according to these templates are systematically available, and because an individual lexical item, though it may in principle be parsable in a number of ways, has associated with it only one particular analysis into constituent Σ (i.e. only one "stress pattern"), we are led to the conclusion that stress foot structure must be associated with a word in the lexicon. A number of quite independent considerations are reviewed which give support to this notion that "word stress" is lexical in English. Among these is the possibility of doing away with the cycle as a mode of application (internal to the word).

To review, it is being proposed that stress feet fall into three basic types, the rough schemata for which are repeated here:

(18) a. 

b. 

c. 

In the following lines I will examine more closely the constraints on just what types of syllables may figure in the various positions of these foot types. I will proceed by examining each of the three major syllable types in English from the point of view of their potential for being "stressed" or "stressless". The three types are the open syllable containing a lax vowel (abbreviated CV̆), the closed syllable containing a lax vowel (abbreviated CV̆C), and the syllable, whether open or closed, that contains a tense vowel (abbreviated CV̄).

A lax open syllable may "bear stress" in initial position, regardless of whether the following syllable is "stressed" or "stressless":

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(19) *Initial "stressed" CV̆*

a. *Before "stressed" syllable*	b. *Before "stressless" syllable*	
*rà*ccóon	*lá*va	*ré*cognìze
*Rà*spútin	*cí*ty	*Mì*ssissíppi
*à*nárthrous	*mó*ra	*mì*santhrópic
*Pàw*túcket	*Klá*math	*À*thabáskan
*sá*tìre	*é*ffort	*Cá*nada
*rá*bbì	*stó*lid	*dí*gital
*ál*lỳ$_N$	*cá*mel	*é*lephant
	*rá*bbit	*pró*minent

But in any other position in the word, "stressed" CV̆ *must* be followed by at least one "stressless" syllable:

(20) *Noninitial "stressed" CV̆*

Òd*é*ssa	Ken*tú*cky	àbraca*dá*bra
Mì*á*mi	bàn*dá*na	àla*bá*ster
m*ỳó*pic	Màr*sá*la	Mìssis*sí*ppi
Hòs*á*nna	va*ní*lla	mìsan*thró*pic
Pàw*tú*cket	de*vé*lop	A*mé*rica
àd*mó*nish	al*ý*ssum	àn*dró*gynous
	Agr*í*ppa	as*tró*nomy

These are the basic generalizations concerning the distribution of "stressed" CV̆ in English words. According to them, then, one does not encounter examples of final "stressed" CV̆, or of a medial "stressed" CV̆ immediately preceding another "stressed" syllable. This appears to be a correct description of the facts. No words of the form [X C V̆] exist in English: *sófă,* but not **sòfá, *sófà; Sénecă,* but not **Sénecà* or **Sènecá.*[10] Moreover, looking at medial syllables, it is noteworthy that one does not find words of a stress pattern like that of *À*dirón*dàcks, Màssa*pé*quòd, e*léc*tròde, La*ér*tès, a*lúm*nì,* for example, where the place of the syllable in roman type is occupied by *CV̆*.

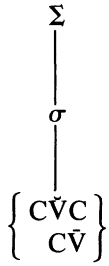
These generalizations can be captured in a straightforward way with the sort of templates being proposed here. What this array of facts suggests is that in all but initial position a CV̆ is prominent only by virtue of being the first (strong) syllable of the bisyllabic stress foot. (Hence the absence of "stressed" CV̆ finally, or medially before "stressed" syllable, which is the start of another Σ). The "stressed" CV̆ of (20) and (19b) are thus all sisters to a σ_w in a stress foot of type (18b). A CV̆ can be a Σ on its own and thereby "independently" prominent only in initial position. The template (18a) for monosyllabic stress feet must be more fully specified to reflect the restrictions on

[10] *a*-final examples are chosen here, because, as was mentioned in section 1, nonlow ω-final vowels are tensed in English, and tense vowel syllables may, as we will see, bear stress in their own right.

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CV̆. The normal state of affairs, as we shall see, is that the monosyllabic Σ may consist of a "heavy" syllable (CV̆C or CV̄), and an additional provision must be made to allow for it to consist of CV̆ in initial position in D_Σ, the footing domain. The elaborated version of (18a) may be formulated as shown in (21).

(21) *The Monosyllabic Stress Foot*

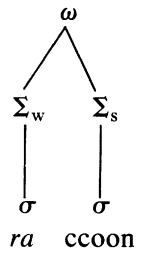
a.[11] 

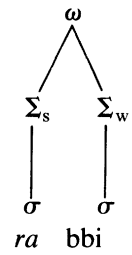
b. initial in D_Σ, 

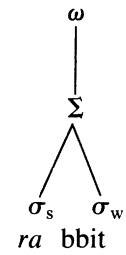
((21b) should be interpreted as saying not that the initial syllable (of whatever kind) in any stem has to be a Σ; rather, it says only that such a syllable *may* be one.) As for the templates (18b) and (18c), we have as yet no motivation for introducing any further specification of the content of the constituent syllables. CV̆ can occupy the position of σ_s in (18b), in which case it is "stressed"; a "stressless" CV̆ is one which occupies the σ_w position in Σ or Σ′.

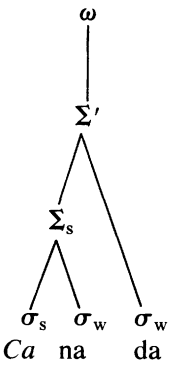
The following examples all conform to the templates as defined so far:

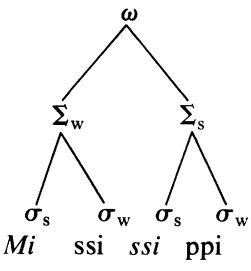
(22)[12]

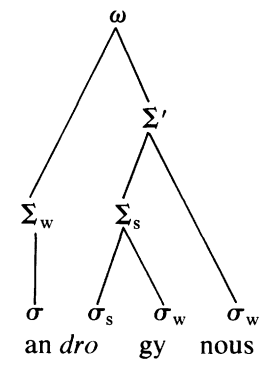










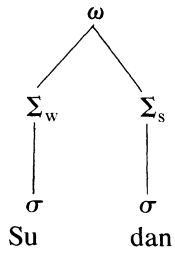


[11] It is not unreasonable to think of tense vowels in English as being bipartite, composed of a vocalic

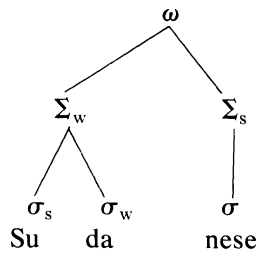
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The CV̆ is footed alone only in initial position (cf. *ràccóon, rábbi*). In the rest of the examples, the "stressed" CV̆ has a "stressless" sister in the Σ. The language's aversion to having a noninitial CV̆ foot is illustrated clearly by the alternation in "stress" pattern of the related words *Sùdán* and *Sùdănése*. The first syllable of *Sudan*, a CV̄, is "stressed"; the second is also "stressed", by virtue of the fact that it is a closed syllable. Thus, the parsing of *Sudan* into stress feet must be (23a). In the derived adjective *Sudanese*, however, this stress foot structure cannot be retained, for the *n* no longer closes the second, lax vowel syllable, which must now necessarily group with another σ (or Σ) within a Σ (or Σ′). In this example, the *da* cannot join with the following syllable, one with a tense vowel, for, as we will see, a CV̄ within a Σ must always occupy the first (strong) position. The *da* therefore foots with what precedes, as in (23b), and is consequently weak, i.e. "unstressed".

(23) a.



b.



To complete this survey of the distribution of "stressed" and "unstressed" lax open syllables, we must examine the case of initial "unstressed" CV̆, such as are found in the *phonetic representations* of the examples of (24):

(24) bălló͡on, hŏrízon, ăttíre, dĭvíne, ăllý$_V$, vănílla, ăróma, Ămérica, pŏlýgamous, etc.

None of the stress foot templates defined so far allows for an initial σ_w, so some elaboration of the analysis is called for. One possible solution would be simply to allow that syllable to "dangle" alone, parsing *vanilla*, for example, as in (25), in underlying as well as surface phonetic representation:

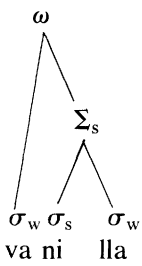
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element and a diphthongal off-glide. If this is the proper treatment of such vowels, the template (21a) can be stated more simply. The restriction on the σ will simply be that "the rhyme of σ branches".

[12] There is an ambiguity of analysis that our templates permit. In certain cases, a bisyllabic sequence may be analyzed either as a bisyllabic Σ, or as a Σ′ with a monosyllabic first Σ. When such ambiguities arise, the Σ analysis will be given as a matter of course. This is why *rabbit*, which could also be analyzed as a Σ′, is given the analysis in (22). Presumably, both analyses would receive the same phonetic interpretation.

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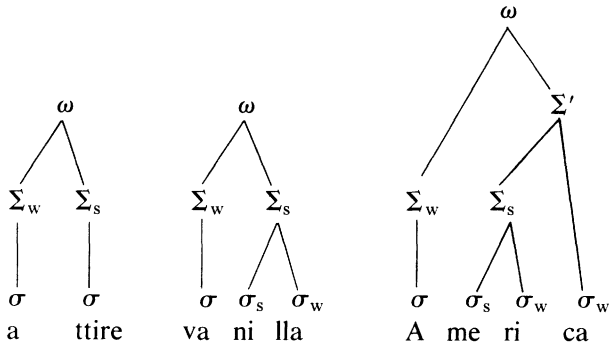
(25)



This solution is undesirable, however, for it seems correct to impose the condition that all syllables are "footed" in underlying representation, that is, at the point where prosodic well-formedness conditions come into play. This condition makes the prediction, a correct one, that "stray" syllables are in general not possible in English.

The correct solution, I believe, involves the contention that the templates proposed here do indeed provide a correct parsing for these examples in *underlying* representation. They would be assigned structures like those in (26), where the initial CV̆ syllables constitute Σ on their own.

(26)



In the surface phonetic representation, however, these syllables cannot constitute Σ, for they are "stressless" and reduced. The mechanism I propose for "destressing" in such cases is a prosodic transformation that will be called Defooting, for its function is to deprive a CV̆ of its Σ status, just in case the Σ is weak. It can be formulated as shown in (27),

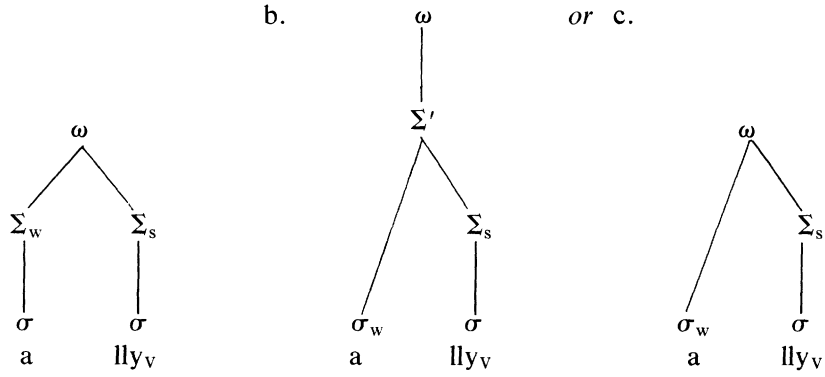
(27) *Defooting*

$$\omega(\Sigma_w(\sigma(C\breve{V})_\sigma)_{\Sigma_w} \ldots)_\omega \Rightarrow \omega(\sigma_w(CV)_{\sigma_w} \ldots)_\omega$$

and will map an underlying representation like (28a) into a derived representation like (28b), or (28c), depending on whether or not one allows an unconventional Σ or unfooted syllables in derived structure. (No position will be taken on this matter here.)

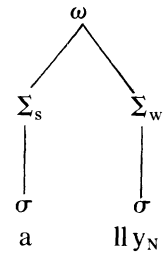
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(28) a. b. *or* c.



In either case, the syllable is now weak with respect to a sister strong and is thus interpreted as nonprominent. It should be noted that Defooting applies only to lax open syllables. This is because neither CV̆C nor CV̄ is stressless in initial position (except when they are subject to a special rule of Latinate Prefix Defooting, cf. LP (p. 288), which will not be discussed here). Also worthy of note is the condition that the Σ dominating the defooted CV̆ be weak, which shows clearly that Defooting must "follow" the determination of prominence relations in ω by the LCPR.[13] Consider the pair *ăllý*$_V$/*állỳ*$_N$. The noun form has the same underlying Σ constituency, (29), as the verb does, cf. (28). Their underlying representations differ only in that the LCPR assigns them different prominence relations. (The stress foot of the noun has an *s/w* pattern, as would be expected, given that the righthand foot does not branch.

(29)



Verbs are subject to a special provision in the LCPR (cf. LP (p. 308)) and so receive rightward prominence even if the righthand Σ does not branch—from whence the *w/s* pattern of *ăllý*$_V$.) The further, surface, difference—the "stresslessness" of the initial CV̆ in *ăllý*$_V$—is introduced by Defooting. At the underlying level, the difference in the *állỳ*$_N$/*ăllý*$_V$ pair is precisely that found in other noun-verb pairs such as *tórmènt/tòrmént, súrvèy/sùrvéy, tránsfèr/tràns fér*, etc.[14] In these later cases, however, the heaviness of

[13] Note that, in contrast, in their treatment of the LCPR, LP essentially build the conditions involved in Defooting into the prominence rule itself.

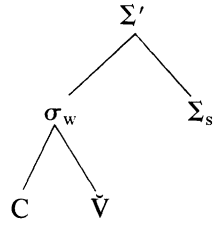
[14] As LP point out, in many instances of such noun/verb pairs, the prefix's Σ_w is destressed (by Prefix Destressing, i.e. Defooting), even if a CV̆C. Cf. *cónvict*$_N$ vs. *cŏnvíct*$_V$.

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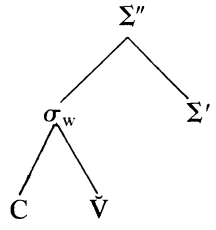
the initial Σ_w in the verb makes it impervious to Defooting, and so an initial "secondary stress" appears phonetically. It is an advantage of this approach, then, that it permits a uniform characterization of these noun-verb pairs at the underlying level. A further advantage of the Defooting approach is that it embodies a claim which is empirically well-founded: that it is normal for an initial weak stress foot containing just $C\check{V}$ to be defooted, i.e. "destressed". The analysis treats *ràccóon*, which maintains initial Σ structure, as an exception: *raccoon* must be marked with the exception feature [−Defooting]. This seems entirely appropriate. (The case of *ràccóon* should be contrasted with forms like *rábbì, Sápphò, cáthòde, Híttìte, éssày, yáhòo,* etc. which, like *álly*$_N$, have a strong Σ dominating the initial $C\check{V}$. These are not exceptional in this analysis.) To conclude these remarks, it should be pointed out that the analysis being offered here of initial "stressless" $C\check{V}$ is more or less the counterpart of the analysis put forth by Halle (1973), and taken up by LP, according to which all initial syllables are first assigned [±stress], and then $C\check{V}$ syllables are "destressed" in particular circumstances by a rule of Initial Destressing. (In the LP version, this Initial Destressing is made subject to the prominence relations of the word, defined in *w/s* terms; the analysis offered here preserves this insight.)

An alternative to the Defooting solution within this framework, one which also conforms to the condition that all syllables be included in a stress foot in underlying representation, would consist in proliferating the number of stress foot types defined for the language, adding two more to the set (18).[15] A first one, (30a), would permit a parsing of *vanilla* in which the initial syllable was footed, cf. (31a); a second, (30b), would be required in order to provide a well-formed parsing of *America*, (31b). The appropriate phonetic interpretations could be assigned directly to the representations of (31), with their σ_w in initial position.

(30) a.



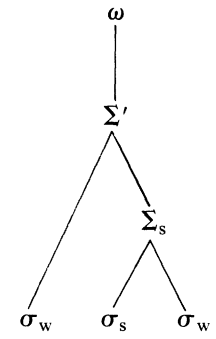
b.



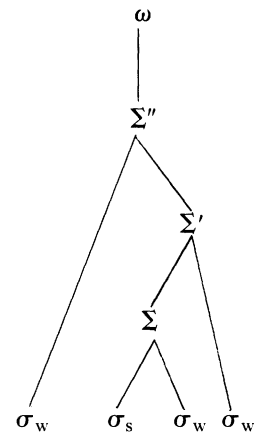
[15] The possibility of such an analysis was pointed out to me by Steve Lapointe.

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(31) a.

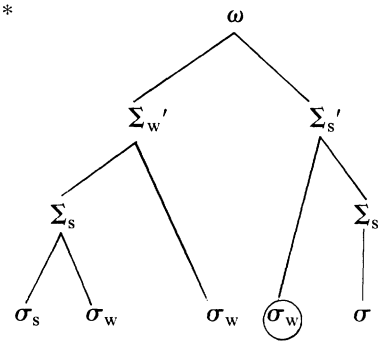


b.



Note that both templates of (30) would have to be restricted to initial position on D_Σ, for if they were permitted internally in D_Σ, they would incorrectly allow for the existence of words containing, for example, a sequence of three weak syllables, as in the hypothetical (32).

(32) *

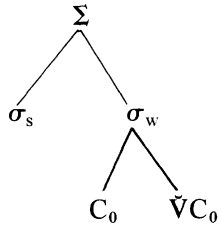


Moreover, the initial syllable position in these new templates would have to be restricted to lax open syllables, since, except in the Latin prefix cases alluded to, heavy syllables are not nonprominent in initial position. An analysis such as this permits one to do away with a prosodic transformation of Defooting, but this is not an obvious advantage, inasmuch as one cannot in general restrict the theory of phonology so as to disallow such rules (cf. Selkirk (forthcoming)). The two analyses, couched within the same general framework, must be evaluated on the basis of their relative descriptive adequacy. On these grounds, the expanded template solution is excluded, for it is neither able to provide a characterization of examples like *ràccóon* as exceptions, nor is it able to attribute to pairs like *ăllý*$_V$/*àllý*$_N$ a relation in their underlying structure that is on a par with that in other noun–verb pairs. We should conclude, then, that the prosodic transformation of Defooting provides the optimal solution to the problem of initial "stressless" CV̆ in English.

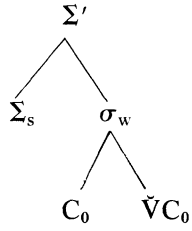
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Turning now to the syllable type $C\bar{V}$, we will see that it has a far more restricted distribution within the stress foot templates than does the lax open $C\breve{V}$. The claim can be made that, in underlying representation, $C\bar{V}$ never appears in the weak syllable positions of Σ and Σ', which is to say that it has to be either the σ_s of the bisyllabic Σ, or the lone syllable of the monosyllabic Σ. The templates (18b) and (18c) will be revised in order to embody this claim:

(33) *The Bisyllabic Stress Foot*



(34) *The Stress Superfoot*

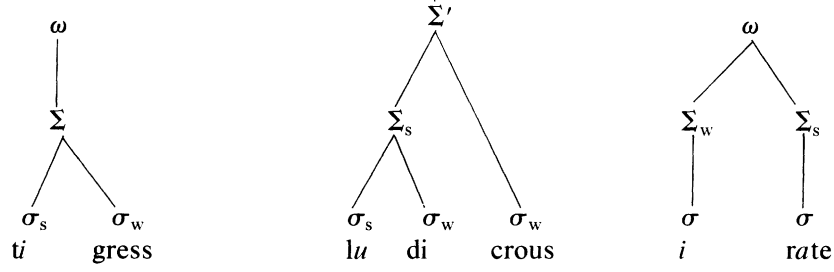


In the context of the present framework this is tantamount to saying that $C\bar{V}$ is always "stressed". A host of forms reflect the claim made by the revised templates:

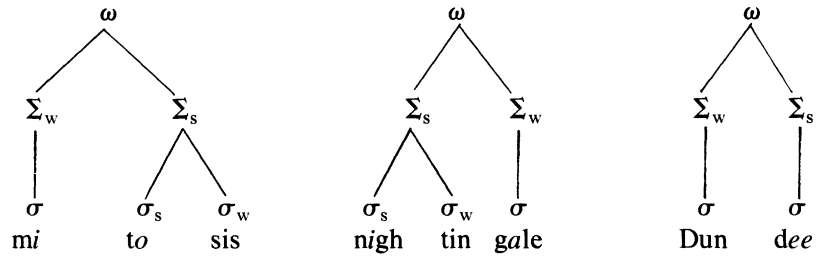
(35) sófa, véto, tígress, phýlum, aróma, horízon, Èlmíra, áileròn, Tàtamagóuchi, móuntainous, glýcogen, lúdicrous, Háverhill [héjvr̩l], nítràte, Péquòd, ìráte, Pèkíng, Tènnessée, níghtingàle, mìtósis

Here are a few representative examples, provided with their prosodic structure analysis.

(36)



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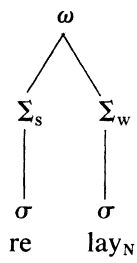
In these instances the surface prosodic structure directly reflects the underlying prosodic structure.

There are certain surface forms which do not conform to our generalization about underlying representation, however, for they include tense vowels which are not in "stressed" syllables, i.e. not in a strong position in a stress foot. These fall into two classes: those at the end of the word (e.g. *cítў, móttŏ, spaghéttĭ, Chicágŏ,* etc.), and those in prevocalic position (e.g. *Canádĭan, ethérĕal*, etc.). The standard analysis of these forms, which I adopt, is that the tensing of the vowels in these contexts is introduced by phonological rules in the derivation and is thus not present in underlying representation (cf. SPE). The first set of cases is taken care of by the rule which tenses all nonlow vowels in ω-final position; the second type of tensing is brought about by a prevocalic tensing rule. Incorporating these rules into a grammar of English allows expression of the generalizations that all prevocalic vowels and nonlow ω-final vowels are [+tense] in English. It also permits our generalization concerning the place of tense vowel syllables in stress foot structure to remain intact. The apparent counterexamples cited here will be analyzed as being underlyingly [−tense].

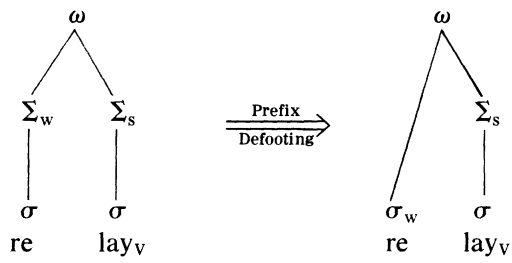
A further sort of case may be thought of as problematic for the claim that every $C\bar{V}$ is in a nonweak position in underlying representation, but again we will see that this turns out not to be so. At issue are vowels which are arguably [+tense] in underlying representation, but which on the surface are in σ_w position and are reduced. The first class of cases involves the underlying tense vowels of the Latinate prefixes *de-*, *re-*, etc. (cf. *défèct*$_N$, *rélày*$_N$). In forms where ω-prominence falls to the right of these prefixes, they may emerge "stressless" and reduced, as in *dĕféctive, rĕláy*$_V$. Our analysis is that underlyingly those prefix $C\bar{V}$ syllables are indeed strong, in these cases being Σ on their own, and that a rule of Prefix Defooting (the counterpart to the LP Prefix Destressing) deprives the prefix of foot-hood, if the Σ itself is weak. Thus, the noun *rélày* is to be contrasted to the verb *rĕláy*, which is subject to this prosodic transformation,

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(37) a.



b.



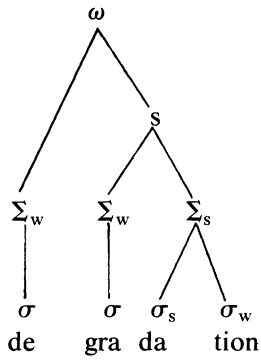
and where the tense vowel contained within the now weakened (i.e. "stressless") syllable is subject to a further process of Detensing (38), and then to the process which reduces lax vowels in weak syllables.

(38) *Detensing*[16]

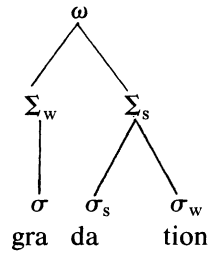
$$\begin{bmatrix} +\text{vocalic} \\ +\text{tense} \end{bmatrix} \rightarrow [-\text{tense}] \,/\, {}_{\sigma_w}(C_0 ___)_{\sigma_w}$$

Given this analysis, then, the prefix examples do not present problems for the generalization concerning the status of CV̄ in underlying representation. The second class of apparently problematic cases contains such forms as *èxplănátion, dègrădátion, dèprĭvátion, rótătòry*, etc., whose medial roman vowels are undeniably [+tense] in underlying representation (cf. *expláin, depríve, degráde, rótàte*), but which are, like the prefix cases, "stressless" and reduced on the surface. Again, it can be argued that in underlying representation the CV̄ syllable constitutes a stress foot, and that in the examples cited, the Σ status of the syllable has been lost through the offices of a prosodic transformation. Consider the derivation of *dègrădátion*. At the outset it has the prosodic structure of (39). The CV̄ syllable of the root *-grad-* is footed here, just as it is in the related form *gradation* (40).

(39)



(40)



[16] Notice that since a tensed vowel in a final weak syllable is never detensed and reduced (cf. *Chicago*), the ω-domain rule of final tensing obviously takes precedence over this σ-domain rule.

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The weak medial CV stress foot, but not the initial one, is evidently subject to a defooting rule (41), and to the further processes of vowel reduction.

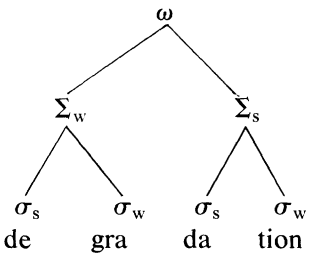
(41) *Medial Defooting*[17]

$_{\omega}(P\ _{\Sigma_w}(_{\sigma}(CV)_{\sigma})_{\Sigma_w}\ Q)_{\omega} \Rightarrow\ _{\omega}(P\ _{\sigma_w}(CV)_{\sigma_w}\ Q)_{\omega}$

Condition: P, Q $\neq \phi$

It seems reasonable to propose that the defooted weak syllable associates with the preceding Σ, as in (42).

(42)



The reason for this is that the syllable "acts as if" it were part of that Σ, and not the one that follows. This is shown by a form like *rótătòry*, where the flapping of the roman *t* indicates (in my analysis) that Resyllabification has taken place, associating the *t* with the preceding syllable. (Recall that Resyllabification operates only within the Σ domain.) To sum up this discussion, then, it has been argued that all CV̄ in English are strong in underlying representation, being either monosyllabic Σ on their own, or the first, strong, member of a bisyllabic Σ. Any apparent divergences from this generalization, which is captured in the revised templates (21), (33), and (34), are the result of the application of well-motivated prosodic transformations and/or rules of segmental phonology.

The last syllable type to be examined is the lax closed syllable CV̆C. The generalization to be made here is that the distribution of CV̆C is entirely free; it may be either "stressed" or "stressless", occupying any of the syllable positions in the three major stress foot types. CV̆C may be a Σ on its own, as in (43a), or the σ_s of a bisyllabic Σ, as in (43b).

(43) a. gýmn*àst*, rédr*èss*, M*àr*sála, N*àn*túcket, *Àt*lántic, *fàndán*go, tróubad*òr*, cátamar*àn*, dèp*òr*tée, *èxpòr*tátion

b. j*ún*ta, h*ún*dred, p*én*dant, el*éc*tric, Àtl*án*tic, Char*ýb*dis, *Úr*sula, p*éc*toral, v*ér*dict

In such cases, it is "stressed". The CV̆C may also be the σ_w of Σ, as in (43c), or the σ_w of Σ', as in (43d):

[17] The LP rule of medial laxing has the same effect as this rule of Medial Defooting.

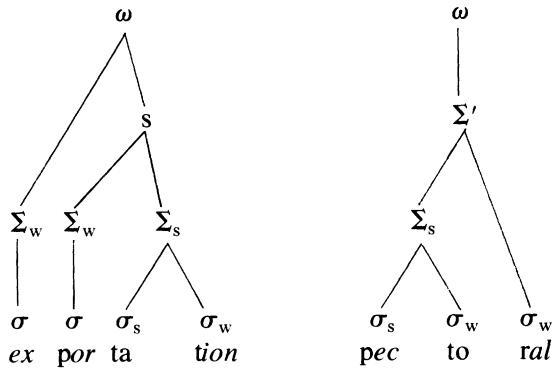
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(43) c. mód*est*, tígr*ess*, húndr*ed*, pénd*ant*, eléctr*ic*, Nàntúck*et*, àn*ec*dóte, Àtlánt*ic*, mìs*an*thróp*ic*, vérd*ict*

d. ánaly*st*, péctor*al*, orígin*al*

These are the "stressless" cases. As formulated, the revised templates (21), (33), and (34) already allow for these possibilities, so no additional revisions need be made. The following examples are illustrative:

(44)



What we have seen thus far is that the generalizations concerning "stress placement" in English words can be summed up in a set of templates defining the stress foot Σ and the stress superfoot Σ'. These templates are considered to be prosodic well-formedness conditions. The claim that is being made is that the sequence of syllables in any word of English can be parsed according to these templates at underlying representation. More precisely, the claim is that all the syllables on the footing domain, D_Σ (contained within the word), must be included in some combination of Σ or Σ' conforming to the templates. Surface divergences from the predicted parsings are introduced in the course of the phonological derivation.

Aside from being desirable on the grounds that it presupposes a simpler, more constrained theory of prosodic structure, the analysis proposed here can be shown to be superior to the standard analysis with its [±stress] assignment insofar as it gives a better account of the facts of English. The stress rule of an analysis of the standard type takes the form of (45) (repeated from (5)), or resembles it rather closely:

(45) $[+\text{syll}] \rightarrow [+\text{stress}] / [\ldots \underline{\quad} C_0 (V(C)) (V\ C_0)]$

The standard interpretation of a rule such as (45) is as follows. If a vowel is (a) in a syllable third from the end of the stem, and (b) the following sequence of segments conforms to the pattern $\underline{\quad} C_0V(C)V\ C_0$,[18] then it receives stress. If conditions (a) and

[18] Actually, the sequence $\underline{\quad} C_0V(C)VC_0$ is a collapsing of two environments: $\underline{\quad} C_0V\ VC_0$ and $\underline{\quad} C_0VCVC_0$, and given the standard interpretation of parentheses the first would be expanded before the other, i.e. would have priority. For the sake of simplicity, we treat these as the same context in the discussion.

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(b) are not met, then a vowel will be stressed if (c) it is in a syllable second from the end and (d) it is followed by a sequence of the type ____ C_0VC_0. If (c) and (d) are not met either, then the vowel of the final syllable is stressed. The requirement is thus that the maximal expansion of the rule (with all its parentheses expressed) has priority. This expansion of the rule applies to a word if its conditions are met, and in this case the other, shorter, expansions do not apply. If the longest is not applicable, then the next longest expansion has priority, and will apply if the word meets its conditions, with the result that the other shorter expansions are inapplicable, and so on. This mode of application, associated with the parentheses notation, is known as disjunctive ordering. LP's version of the standard analysis represents (45) as an iterative rule, beginning on the right and moving leftward. It is now recognized that the generalization about English incorporated in a stress rule such as (45), interpreted in the fashion described, is empirically unfounded. In particular, it has been shown that the condition on maximality is not in general met (cf. Ross (1972), Halle (1973)). We have seen many instances of nonmaximality, *vanílla* (not **vánilla*), *pálimpsèst* (not **palímpsest*), *gýmnàst* (not **gýmnast*), *Tìcònderóga* (not **Tìconderóga*) among them. This state of affairs has led advocates of this type of stress rule to allow for the possibility that words will be marked as exceptions to one or another expansion of the stress rule. *Vanílla* would be an exception to the fullest expansion of the rule, and so would go by the second case; *pàlimpsést* (at least its final syllable) would be an exception to both expansions, and so the final syllable would be stressed. Needless to say, the systematic use of such exceptions, such as is required in a full treatment of the English vocabulary, vitiates the claim about maximality that the standard interpretation of (45) embodies. (It should also be pointed out that once maximality is no longer at play, directionality has no role either.) This is one reason that an analysis based on (45) and its horde of exception-feature-marked words should be rejected. The second reason is that this revised sort of stress rule analysis fails to capture certain generalizations which *are* true of English stress patterns. I have in mind the fact that CV̆ is never stressed finally, or before another stressed syllable. The stress rule-cum-exception feature analysis cannot express this fact, since it should be possible for a word like *Séneca*, for example, to be marked as an exception to the two first expansions of the stress rule and so, like *pálimpsèst*, to receive stress on the final syllable. The conclusion is that the particular template analysis that has been articulated here for English is superior to the conventional [±stress] analysis.

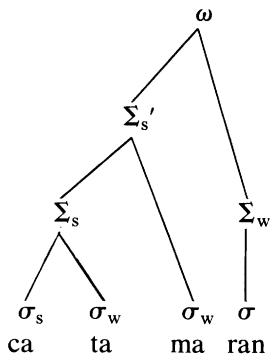
Note that the claim is not being made that the notions of maximality and directionality have no place in the theory of prosodic structure. Clearly this is not so, as McCarthy's (1979) analysis of Arabic (with its left to right foot formation) or HV's analysis of Latin and its maximal feet demonstrate. The claim is merely that in English no choice of one analysis into Σ over another is made on the basis of maximality, or directionality; that is, given certain combinations of syllables, it cannot be predicted which of a (limited) number of parsings into Σ and Σ' will be chosen. (This by no means

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means that "stress" is unpredictable in English, only that in certain cases options are made available.)

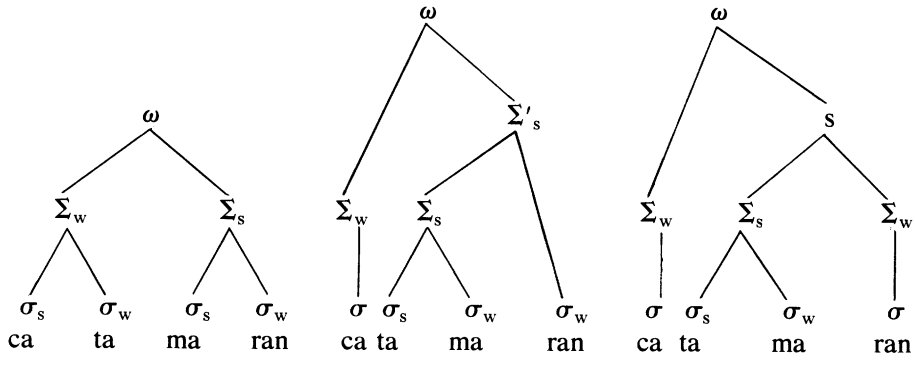
A "choice" between two (or more) analyses in stress feet rarely presents itself in practice, for a particular word (usually) has just one analysis, i.e. just one "stress pattern", and if we know a word we know its Σ analysis. The point is that there exist words with a particular "stress pattern" which could, in principle, have had another. Consider the word *cátamaràn* [kʰǽrəməræn], which will be assumed to contain all lax vowels underlyingly. Given its "stress pattern," its analysis into Σ and Σ′ must be as shown in (46).

(46)



The same sequence of syllables could have been organized into three other patterns, given our templates—*càtamáran* (cf. *Àthabáskan*), *catámaran* (cf. *América*n), *catámaràn* (cf. *Decámeròn*)—as in (47).

(47)



Speakers are usually made aware of such options only when confronted with the written form of new words which they have not heard before. Sometimes the orthography allows for more than one interpretation. "Mistakes" I have myself made, until enlightened, are pronouncing *Mŭlligatáwny* for *Mùlligatáwny*, *Óregòn* for *Óregŏn* [ɔ́rəgən], and *Willamétte* for *Wĭllámĕtte* [wɪlǽmət]. (The last two are the pronunciations of natives

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of Oregon.[19]) The templates allow both the correct and the erroneous pronunciations: they simply involve combining the component CV̆ and CV̆C syllables in different ways.

What I am asserting is that there is nothing in the prosodic structure analysis of English which implies a choice of one option over another. As we will see in the discussion of the cycle, below, morphological factors may interact with prosodic factors and so limit the range of options available in morphologically complex items. But in monomorphemic items, like those just examined, the choice is quite free. The prosodic structure analysis of a lexical item is thus potentially idiosyncratic. The conclusion to be drawn from this is that prosodic structure at the level of the prosodic word and below (including stress feet and syllables) forms part of the lexical representation of a word. Note that the conclusion that "stress" is "lexical" is not forced upon us by prosodic structure theory, only by the facts: in the LP framework the conclusion would have to be that [±stress] designations are in the lexicon (with the possibility in principle remaining open that the suprasegmental structure could be introduced later, in the course of a phonological derivation). In our theory, "stress" has no existence independent from prosodic structure, and so it is the structure itself which is lexicalized.

This conclusion involves a slight modification in the position taken thus far on the point at which prosodic well-formedness conditions are applicable in the grammar. Earlier, it was hypothesized that, universally, the definition of prosodic categories constituted the mapping from syntax (surface structure) to phonology (underlying phonological representation). This may indeed still be correct for a language like French, where word "stress" is absolutely regular. In the grammar of English, however, it is only the categories above the level of prosodic word, like phonological phrase and intonational phrase (cf. Selkirk (1978a; 1978b; forthcoming)), which are defined at this point in the derivation of a sentence. If "word stress" is lexical, then prosodic structure below ω already forms part of the syntactic surface structure, along with the segments which have traditionally been seen as forming part of a lexical entry. Furthermore, a modification must be made in our conception of the *role* of the prosodic well-formedness conditions, for σ, Σ and ω in the grammar of English. They will now have to be seen as lexical redundancy conditions, part of the speaker's knowledge of the grammar, to be sure, but a part which plays no role in a derivation.

Facts relating to word formation can be cast in a new, clearer, light, given the assumption of lexical stress. For example, Siegel (1974) has pointed out that certain processes of affixation are dependent on the stress pattern of the root to which they attach. Among these is the suffixation of the *-al* which forms nouns from verbs (*perúse/perúsal*). This *-al* attaches only to verbs with final stress. That such restrictions should in fact exist is predicted by the assumption of lexical stress, given, in addition, the assumption—one argued for by Aronoff (1976) and adopted by me (cf. Selkirk (forth-

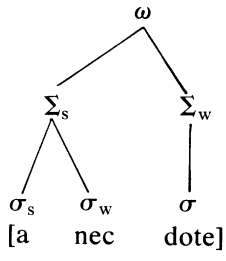
[19] I am grateful to Ann Jones, a native Oregonian, for pointing out how the last two place names are "really" pronounced.

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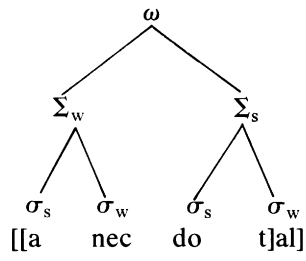
coming))—that word formation is in general defined in terms of existing lexical items, meaning that a new lexical item is constructed on the basis of an old one. Further support for seeing "stress" as forming part of the lexical representation of a word is provided by the analysis of psycholinguistic data. Tip of the tongue phenomena (cf. Brown and McNeill (1966)), malapropisms (cf. Fay and Cutler (1977)), recall experiments (Robinson (1977)), lexical access experiments (Engdahl (1978)), and more, all attest to the fact that word "stress" is an important organizing principle for the lexicon.

There is an interesting, nontrivial, consequence of having "stress" (= prosodic structure) associated with an item in the lexicon: the cycle, as a mode of rule application internal to the word, can be done away with. Words, as we all know, have internal morphological structure, which can be represented as a labeled bracketing. So *anecdotal* has the analysis [[anecdot]al], and *condensation* [[condens]ation], *compensation* [[compensat]ion] (leaving the labels of the brackets aside). Now it has been widely observed that the "stress pattern" of morphologically complex items like these must be seen as resulting from the "stress pattern" of the embedded item in combination with the particular "stress"-related properties of the affix. In the noun *ànĕcdóte*, the medial CV̆C is stressless and the initial CV̆ is stressed. In *ànĕcdótal*, this stressing is preserved—though, given the syllable composition of the form, another pattern, e.g. *ănècdótal*, would have been possible. Thus, the adjective reflects the stress properties of the parent form. (The *-al* has no peculiar properties of its own, except that it is "stressless".) To take another example, the contrast in the stressing of the medial CV̆C of *còndènsátion* vs. *còmpĕnsátion* can be attributed to the fact that the former is derived from the verb *cŏndénse*, where the syllable in question bears stress, while the latter derives from *cómpĕnsàte*, where the syllable lacks stress. The cycle was proposed (in SPE) as the principle which made sense out of such facts. According to the principle of the cycle, the stress rule would apply first on the innermost cyclic domain, and then move on to the next higher domain, and so on. In this way, the properties established on the inner domain always made a contribution to the overall stress pattern of the word. Clearly, though, if stress is part of the representation of a lexical item, there is no motivation for a cyclic application of this sort. *Anecdote* has its own representation in the lexicon, as does *anecdotal*—stress is not assigned by rule in a derivation. In prosodic structure terms, the representations of the two are (48a) and (48b), respectively.

(48) a.



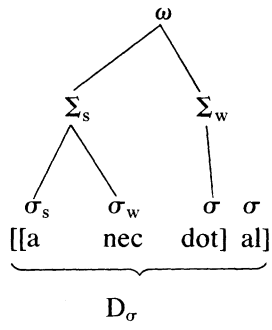
b.



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The close relation between the stress patterns (i.e. prosodic structures) of the two items would have been established, or laid down, at the point (in time) when *anecdotal* was made from *anecdote*, as a result of the process of word formation. In terms of prosodic structure theory, the suffixation of *-al* can be thought of as setting up a situation in which *anecdote*, already supplied with prosodic structure, is now accompanied by an additional syllable within the same footing domain, D_{Σ}. The structure can be visualized as in (49).

(49)

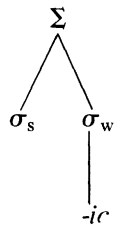


This newly-formed word is subject to the conditions of prosodic well-formedness, and insofar as all syllables must be included in some stress foot (and all feet in a prosodic word) on that domain, a restructuring is necessary, in order that a well-formed prosodic structure be obtained. Such a restructuring can be thought of as coming about automatically. It is never wholesale, and apparently only introduces the *minimum* change that is required in order that the new whole be well-formed. In the case of *anecdotal*, this will amount to footing *-al* with the preceding Σ dominating *-dote*, which has the consequence of drawing the *s* of the ω onto the now-branching righthand Σ, by the LCPR. Our approach to the facts motivating the cycle is thus to see them as resulting from submitting a derived word, formed from a prosodically structured lexical item plus an affix, to the prosodic well-formedness conditions of the language, constrained to bring about only minimal modifications in the original structure. I should mention in passing that it is implied by this theory, which gives no place to the cycle, that, in a language where stress has no exceptions and hence need not be lexical, there will not be stress patterns reflecting the internal morphological composition of words.

Certain affixes have stress-related properties which are rather more specific than those of *-al*. Though there is not room here to explore them all (the reader is referred to LP (pp. 274–279) and to Selkirk (forthcoming) for some treatment of this topic), it will be useful to examine a couple of cases, to make clear the general lines of the analysis that will be pursued. Consider *-ic*, for instance. It is well known that *-ic* requires that the syllable preceding it be stressed. In prosodic structure terms, this means that *-ic*, stressless itself, insists on being the second element of a Σ:

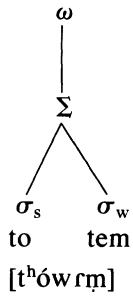
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(50)

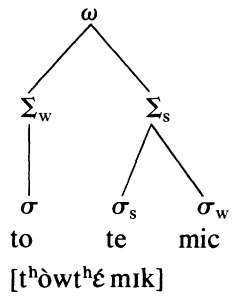


Meeting this restriction may involve a not insignificant change in the stress foot structure of the root of a derived word, as shown in (51):

(51) a.

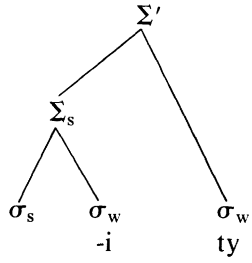


b.



The general constraint governing restructuring must thus be that only minimal changes in prosodic structure are made, unless affix-particular conditions specify otherwise. Consider next the affix *-ity*. It consists of two syllables, yet these do not themselves constitute a Σ—the *i* is not "stressed". They must be seen as forming necessarily the last two σ_w of a stress superfoot:

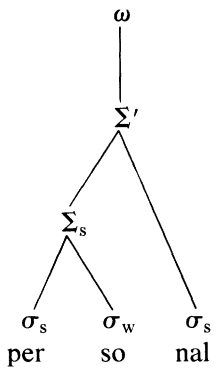
(52)



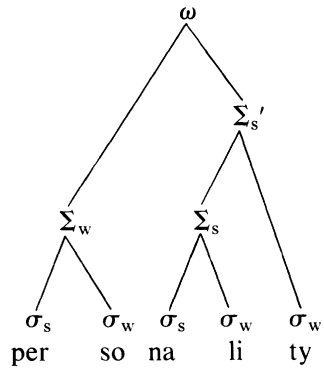
These observations account for the alternations in stress pattern in *pérsonal/pèrsonálity*, or *specífic/spècifícity*, for example, as in (53) and (54).

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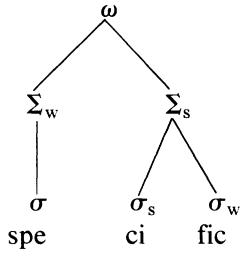
(53) a.



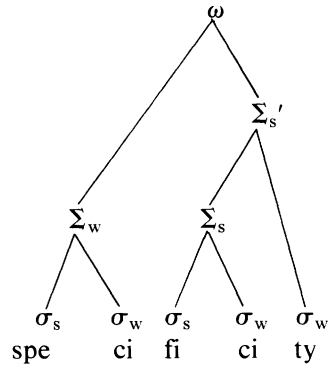
b.



(54) a.

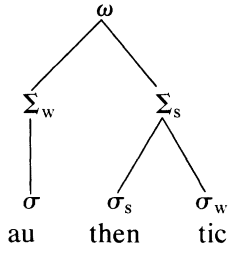


b.

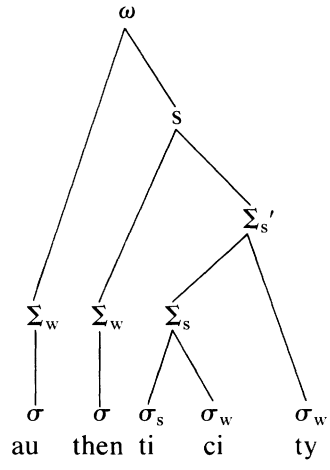


The modification in Σ in the second example is worthy of note. Because the syllable with *-ic* is required to join with the following *-ity* in a Σ, it abandons its former footmate *-ci-*, which, being CV̆, cannot stand alone as a Σ and must thus attach to the preceding Σ, *spe-*. It is therefore "unstressed". The case of *specificity* should be contrasted with one like *aùthèntícity* (and its parent *aùthéntic*); cf. (55), where the medial *then*, as a CV̆C, can, and does, remain a Σ, bearing "secondary stress".

(55) a.



b.

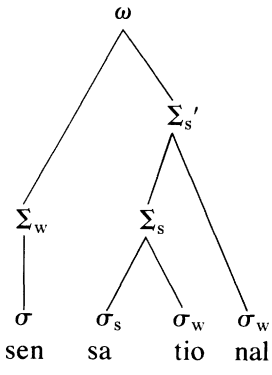


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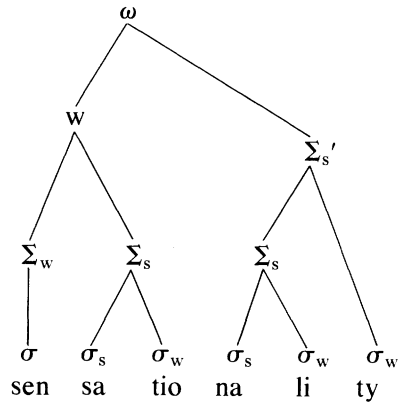
These examples illustrate, then, that morphologically governed modifications in the prosodic structure of derived forms may, in conjunction with prosodic well-formedness conditions, have nonlocal consequences of some significance. These examples also illustrate the fact that the stress properties of an outer affix take precedence over those of an inner one: the need of *-ity* to take on the preceding syllable as a strong head overrides the need of the *-ic* itself to be the σ_w with a preceding sister σ_s in a Σ. It might be thought that the cycle could explain such facts—the morpheme on the outermost cyclic domain would naturally have an effect on the morphemes further in, but not vice versa. In the context of our cycle-less theory, these facts will be seen as reflecting an order in word formation. The *-ic* is affixed and its prosodic effects felt "before" the affixation of *-ity*, which "later" imposes its own prosodic restrictions.

Kiparsky (1979) addresses the question of the cycle in the framework of the LP theory of stress. The central argument made by Kiparsky for a cyclic assignment of "metrical" structure in English words can be reinterpreted now as being an argument to the effect that the prominence relations within the prosodic word, i.e. those assigned by the LCPR, must form part of the lexical representation of an item. Kiparsky argues that the presence of secondary (vs. tertiary) stress in a form like $\overset{3}{s}\overset{2}{e}\overset{1}{nsationality}$, which is derived from $\overset{2}{s}\overset{1}{ensational}$, is to be attributed to the fact that in the base form that syllable was primary-stressed (i.e. bore the *s* to its partner's *w*). In our terms this would simply mean that the prosodic word structure and prominence relations of *sensational* (56a) were carried over, to the extent compatible with well-formedness, in the derived form (56b).

(56) a.



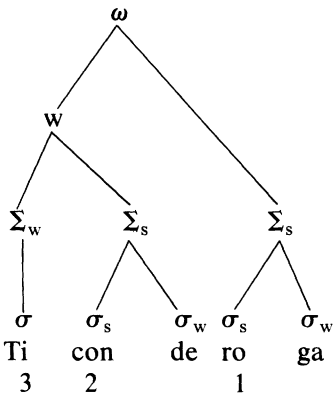
b.



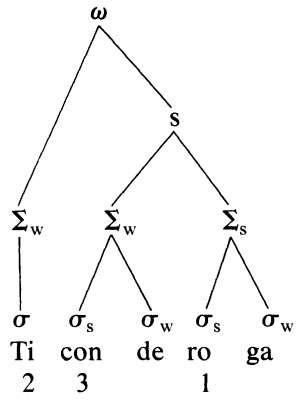
It should be noted here that the supra-Σ branching structure in (56b) is not right-branching, which is to say that it does not conform to the well-formedness conditions on ω, as defined. In point of fact, Kiparsky denies that right branching is a generalization for ω in English, citing two alternate pronunciations of words like *Ticonderoga* (57) in support of this contention.

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(57) a.



b.



Indeed, it is the availability of these two pronunciations for a monomorphemic form, but the limitation to just one (with the morphologically-determined left branching) in the case of *sensationality*, which forms the substance of his argument that *sensational*'s structure and prominence pattern is being preserved in the derived form. There is unfortunately not the space here to investigate the implications of Kiparsky's claim about the nature of the branching in the ω in English, but I would like to say that in dropping wholesale the condition that right branching is at least preferred, the variety of generalizations outlined by LP concerning the location of primary stress in English words go uncaptured.

As a last note on Kiparsky's claims for the cycle, it should be pointed out that in his discussion of the Rhythm Rule, the ordering (58) which he argues for does not, if correct, provide evidence for the cyclic application of phonological rules on successively less embedded morphological structure domains within the word.

(58)
- Rhythm (in the word)
- Initial Destressing (a "word-level rule")
- Rhythm (on the phrase)

At issue here is the claim that the Rhythm Rule, a prosodic structure transformation, applies at the level of the word (prior to Initial Destressing) before it applies to sequences larger than the word. In Selkirk (1978b), I argue that the above-the-word-level domain of the Rhythm Rule is properly construed as the phonological phrase (φ), a unit of prosodic structure, not of syntactic structure. I would argue that, if indeed the same process of Rhythm is applying at the level of the word, the domain, properly defined, is the prosodic word, ω. If it holds, then, Kiparsky's argument is an argument for the possibility that prosodic structure transformations may cycle on domains defined in terms of prosodic categories. It is not an argument in favor of the standard position that phonological rules cycle on domains defined in terms of syntactic, or morphological, structure.

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This brings us to the end of our examination of English stress within the framework of a theory of prosodic structure. The main goal of this section has been to provide an analysis of "stress placement" which captures all the known generalizations, but which does not make appeal to the feature [±stress]. I hope this goal has been met to the reader's satisfaction. Certain other important questions concerning the stress of English words have only been touched on here. These would have to be looked at in some detail, were the new prosodic structure analysis being presented here to be complete. The Lexical Category Prominence Rule, for example, demands further scrutiny—not only because of the observations made by Kiparsky, but also because, within the present framework, its role is confined to the ω. (It does not determine the prominence relations within the Σ, as LP had envisioned.) Moreover, the stress-related properties of particular affixes merit a more detailed treatment. (The reader is referred to LP's discussion of the various morphologically-governed "modes of retraction".) And the issue of the syntactic domain of the formation of σ, Σ, and ω cries out for fuller discussion. In Selkirk (forthcoming) I attempt to answer to these needs. This is done in the context of a rather elaborate defense and illustration of the theory of prosodic structure.

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Department of Linguistics
South College
University of Massachusetts
Amherst, Massachusetts 01003

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